



Swami Keshvanand Institute of Technology, Management
& Gramothan. Ramnagar, Jagatpura, Jaipur
Department of Computer Science & Engineering
Project (2026-27)

CSE-53

Functional Requirements Finalization

Project ID: SKIT/CSE/2023-2027/53..... Branch: Computer Science Section: D

Project Title: Smart Pothole Detection System.....

SDG Mapping: Sustainable Cities and Communities.....

Project Overview:

Potholes are currently identified through slow manual surveys or citizen-complaint apps, both of which are inconsistent and leave much of the road network uncovered, delaying repairs. The system uses a vehicle-mounted camera and a deep learning model to detect potholes, tag their GPS location, store the results in a database, and generate a quantitative road-condition score.

Objectives: Main objectives of the project

- 1) To collect and preprocess a labeled pothole image dataset suitable for model training.
- 2) To develop and fine-tune a YOLOv8-based deep learning model for accurate pothole detection in road video frames.
- 3) To geotag each detected pothole using GPS data synchronized with the recorded video.
- 4) To compute a quantitative road-condition score for each road segment based on pothole frequency, size, confidence.
- 5) To build an interactive dashboard that visualizes detections and road-condition scores on a map.

Functional Requirements:

Detail the specific functional requirements of the project.

ID	Requirement Description	Priority (High/Medium/Low)
FR-001	System shall accept a road video file recorded via smartphone as input and validate that the uploaded video is in a supported format (MP4).	High
FR-002	System shall accept / parse a GPS log file synced with the video recording (Excel).	High
FR-003	System shall display an interactive map with all geotagged detections and allow the user to click a detection markers to view (image, confidence, time).	High
FR-004	System shall display summary statistics (total potholes detected, average road condition score, and average potholes per road segment).	High
FR-005	System shall allow adjusting the frame-extraction interval (e.g., every 25-30 frames) as a configurable parameter and discard detection below minimum confidence threshold to reduce false positives.	Medium
FR-006	The system shall process a 1-minute road video and produce detections within 2 minutes on the specified (min. 8GB RAM) laptop hardware.	Low
FR-007	System shall store each detections record (frame-reference, GPS location, confidence score, timestamp) in a SQL database.	High

Non- Functional Requirements:

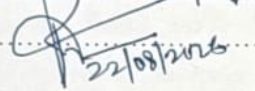
Detail the specific functional requirements of the project.

ID	Requirement Description	Priority (High/Medium/Low)
NFR-001	The dashboard map shall render within a reasonable time even with hundreds of markers.	Medium
NFR-002	The dashboard shall provide clear error messages if a video upload or processing step fails.	Medium
NFR-003	Code shall be organized into separate modules (detection, GPS-matching, scoring, dashboard) for independent updates.	Medium
NFR-004	The potholes detection model shall achieve a minimum accuracy of 75% MAP and IoU. threshold of 0.5 on the test dataset.	High

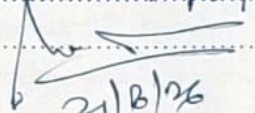
Assumptions and Constraints

- 1) Assumption 1... The vehicle-mounted camera provides sufficiently stable and clear video footage for detection.
- 2) Assumption 2... GPS signal is available and reasonably accurate (within a few metres) throughout video recording.
- 3) Constraint 1... Model training and inference are limited by the available laptop hardware (minimum 8GB RAM, no dedicated GPU).
- 4) Constraint 2... The project timeline is fixed from 10 August 2026 to 6 March 2027, which bound the achievable dataset size and feature scope.

Project Mentor

Name Pawan Kumar Patidar
Designation Assistant Professor
Signature  22/08/26

Lab Coordinator

Name Manish Bhardwaj
Designation Assistant professor
Signature  21/8/26